Joanna Kaleta

PhD Researcher in Computer Vision, Neural Rendering & Generative Models

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RESEARCH PROFILE

PhD researcher in 3D computer vision and generative modeling. I am interested in combining 3D spatial representations with generative models to build methods for reconstructing, understanding, and generating spatially faithful 3D scenes. My work spans feed-forward 3D reconstruction, Gaussian Splatting, neural rendering, diffusion models, and recently video-based scene representations.

RESEARCH EXPERIENCE

08/2026 - Present — **Simon Fraser University, GrUVi Lab - Visiting PhD Student**
Host: Prof. Andrea Tagliasacchi
- Investigation of spatial understanding in video diffusion models: can a video model re-generate a dynamic video, conditioned on photos of the same scene after spatial changes?

02/2026 - 05/2026 — **ETH Zurich, Computer Vision and Geometry (CVG) Group - Visiting PhD Student**
Host: Prof. Marc Pollefeys
- Test-time refinement of feed-forward 3D reconstruction: recurrent refinement of camera poses and Gaussians. We show that pose refinement is an important factor in boosting reconstruction quality.

09/2023 - Present — **Warsaw University of Technology - PhD Student, Doctoral School**
Supervisors: Prof. Tomasz Trzciński, Dr Marek Kowalski
- 3D reconstruction and neural rendering (Gaussian Splatting, Inverse Rendering), generative vision models.

10/2022 - Present — **Sano Centre for Computational Personalized Medicine - PhD Student Researcher**
- Computer vision and generative modeling for medical imaging, endoscopy, and robotics-oriented applications.

SELECTED PROJECTS AND PUBLICATIONS

My full publication and project list, including works under review, is available on Google Scholar and GitHub.

ACM Multimedia 2026 — **AnyStyle: Single-Pass Multimodal Stylization for 3D Gaussian Splatting**
J. Kaleta, B. Świrta, K. Kania, T. Trzciński, P. Spurek, M. Kowalski
Feed-forward, pose-free 3D reconstruction and zero-shot stylization framework with text and image conditioning.

CVPR 2026 Highlight — **LumiMotion: Improving Gaussian Relighting with Scene Dynamics**
J. Kaleta*, P. Wójcik*, K. Marzol, T. Trzciński, K. Kania, M. Kowalski
Leveraging surface motion as additional signal to better disentangle material properties and global lighting in Gaussian Splatting based reconstruction.

WACV 2025 Oral — **LumiGauss: Relightable Gaussian Splatting in the Wild**
J. Kaleta, K. Kania, T. Trzciński, M. Kowalski
Relightable Gaussian Splatting from unconstrained photo collection using Precomputed Radiance Transfer.

MICCAI 2025 — **PR-ENDO: Physically Based Relightable Gaussian Splatting for Endoscopy**
J. Kaleta*, W. Smolak-Dyżewska*, D. Malarz, D. Dall'Alba, P. Korzeniowski, P. Spurek
Relightable Gaussian Splatting for colonoscopy, leveraging camera and light source co-location.

NeurIPS 2024 — **D-MiSo: Editing Dynamic 3D Scenes Using Multi-Gaussians Soup**
J. Waczyńska, P. Borycki, **J. Kaleta**, S. Tadeja, P. Spurek
Hierarchical and Editable Gaussian Splatting representation of Dynamic scenes.

NVIDIA Best Student Paper Award — **Minimal Data Requirement for Realistic Endoscopic Image Generation with Stable Diffusion**
J. Kaleta, D. Dall'Alba, S. Płotka, P. Korzeniowski
NeurIPS Workshop on Diffusion Models 2023
Medical Sim to Real: generate fully annotated realistic endoscopic data based on renders from medical simulator.

INDUSTRY & ENGINEERING EXPERIENCE

05/2022 - 09/2022 — **Appsilon - Machine Learning Engineer**
Deep learning models for image classification and object detection in Data4Good wildlife-protection projects. Developing GAN model for human motion style transfer.

10/2020 - 04/2022 — **RedCompass Labs - Data Scientist**
Anomaly detection, graph analysis and NLP on financial transactions to detect modern slavery patterns; Spark pipelines and synthetic transaction data generation.

EDUCATION

- **PhD in Computer Science:** Warsaw University of Technology, Doctoral School (2023 - Present).
- **MSc Computer Science:** Warsaw University of Technology - graduated with distinction;
- **Languages:** Polish (native), English (C2)

TECHNICAL SKILLS

Research areas: 2D/3D computer vision, neural rendering, inverse rendering, synthetic data generation, medical imaging, medical Sim-to-Real

Methods: feed-forward 3D reconstruction with geometry transformers, Gaussian Splatting, diffusion models, deep learning models

Programming tools: Python, PyTorch, distributed training, SLURM, Blender, COLMAP; Linux, Git, Docker; AI-assisted development (Cursor, Claude Code)